Dual-Path Deep Learning for B2B Sales Outcome Prediction: Practical Pipeline with Failure Analysis and Explainability

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***Abstract*—This report presents an IEEE-style project summary for B2B sales outcome prediction on heterogeneous tabular CRM data. We compare a baseline Logistic Regression model, an advanced machine learning SVM (RBF) model, and a primary deep learning model using a custom dual-path architecture. The dual-path model processes numeric and categorical features separately before fusion, introducing an inductive bias suited to mixed-type tabular data. In addition to predictive metrics, the report emphasizes failure analysis (false positives and false negatives) and SHAP-based interpretability for technical validation and business alignment.**

***Index Terms*—B2B sales prediction, heterogeneous tabular data, dual-path MLP, failure analysis, SHAP, CRM analytics**

I. Introduction

Sales pipeline prioritization requires identifying which opportunities are likely to be won. This project predicts binary deal outcome ($Y \in \{0, 1\}$) and focuses not only on prediction quality but also on practical error diagnostics. The key design choice is a **dual-path deep learning architecture** where numeric and categorical signals are modeled separately before representation fusion.

II. Problem Setup and Data Characteristics

The dataset is heterogeneous and contains both numeric and categorical fields. Let

$$\mathbf{x} = [\mathbf{x}_{num}, \mathbf{x}_{cat}], \quad y \in \{0, 1\}. \tag{1}$$

Why preprocessing is required:

- Numeric scaling (`StandardScaler`) stabilizes optimization and keeps gradients well-conditioned.
- Categorical encoding (`OneHotEncoder`) converts symbolic values into machine-consumable vectors.
- Log transforms (e.g., revenue and employee count) reduce skew and stabilize heavy-tailed distributions.

Regularization rationale:

- L2 regularization ($\alpha = 0.001$) constrains weight magnitude and improves generalization.
- Dropout is not implemented, but conceptually serves a similar anti-overfitting purpose by reducing co-adaptation.

III. Literature Review

A. Category A: Classical ML for Sales Prediction

Paper A1: *Sales Opportunity Win Propensity Prediction* [1].
Paper A2: *CRM Sales Prediction Using Machine Learning* [2].
These works show that classical models are strong practical baselines for structured CRM signals. They are generally easier to deploy, easier to monitor, and often competitive on medium-scale tabular data.

Limitations observed in this category:

- Feature interactions may require manual engineering.
- Nonlinear and cross-type interactions can remain under-modeled.
- Error interpretation is often metric-centric, with limited deep failure diagnostics.

B. Category B: Neural Networks for Tabular Data

Paper B1: TabNet [3].
Paper B2: Deep Learning for Tabular Data Survey [4].
These works motivate architecture-level inductive biases for tabular learning and highlight when deep learning can outperform classical methods, especially with richer interaction learning.

Limitations observed in this category:

- Higher implementation complexity and hyperparameter sensitivity.
- Explainability and stability concerns in operational settings.
- Gains are not guaranteed on every tabular dataset.

C. Positioning of This Project

This project is positioned as a **simple but practical extension**: it uses a custom dual-path DL model to encode heterogeneity while maintaining deployment-friendly analysis artifacts (model-tagged reports, confusion matrices, high-confidence error CSVs, and SHAP summaries). The central contribution is failure analysis for decision support, not just leaderboard optimization.

IV. Methodology

A. Model Set

- Baseline: Logistic Regression
- AML benchmark: SVM with RBF kernel
- Primary DL: Custom Dual-Path MLP

B. Dual-Path Architecture Logic

Numeric and categorical branches are learned separately:

$$\mathbf{h}_{num} = \mathrm{ReLU}(W_{num}\mathbf{x}_{num} + b_{num}), \tag{2}$$

$$\mathbf{h}_{cat} = \mathrm{ReLU}(W_{cat}\mathbf{x}_{cat} + b_{cat}), \tag{3}$$

then fused:

$$\mathbf{h}_{fuse} = [\mathbf{h}_{num}; \mathbf{h}_{cat}], \tag{4}$$

and passed through final dense layers to produce

$$\hat{y} = \sigma(W_o\mathbf{h}_2 + b_o). \tag{5}$$

Inductive bias: numeric trends and categorical identity patterns are different statistical objects, so learning them in separate paths before fusion can improve representation quality for heterogeneous tabular data.

C. Training Objective

Binary cross-entropy with L2 regularization:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\Big[y_i\log(\hat{y}_i)+(1-y_i)\log(1-\hat{y}_i)\Big]+\alpha\|W\|_2^2. \tag{6}$$

Optimization uses Adam with early stopping on validation loss.

V. Results

From the executed pipeline:
Dual-Path MLP (Primary DL): F1 = 0.7660, Precision = 0.6317, Recall = 0.9729, ROC-AUC = 0.5206.
SVM (RBF): F1 = 0.5173, ROC-AUC = 0.4881.
Logistic Baseline: F1 = 0.4728, ROC-AUC = 0.5140.

Interpretation:

- The primary DL model is recall-oriented and captures most positive outcomes.
- ROC-AUC remains near-random, indicating weak global ranking ability.
- This motivates operational use with threshold control and explicit failure analysis.

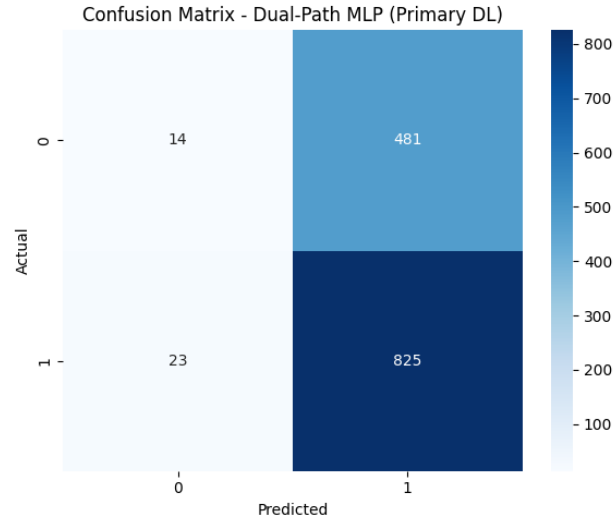

Fig. 1. Confusion matrix of the primary dual-path DL model at threshold 0.50.

VI. Failure Analysis and Business Relevance

The project emphasizes:
False Positives (bad bets) and **False Negatives (missed opportunities)** with high-confidence error extraction segmented by sector, product, and price band.

This converts model output into actionable sales strategy diagnostics. The practical thesis is:

> High-recall prediction is useful only when paired with structured failure analysis that controls cost of bad bets.

VII. Explainability Validation (SHAP)

SHAP is applied through a simplified explainability model on processed tabular features to verify feature-impact consistency with business logic.

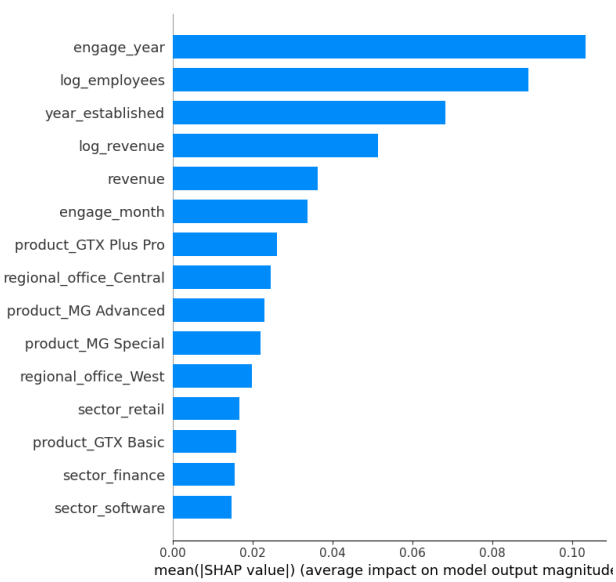

Fig. 2. SHAP global feature importance (simplified explainability model).

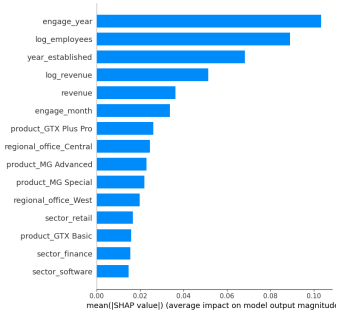

Fig. 3. Enter Caption

Top SHAP signals include engagement time features, transformed company scale variables, and product/region indicators, which are plausible business drivers for deal outcomes.

VIII. Theoretical Notes

A. Gradient Descent Intuition

Gradients indicate local sensitivity of loss to parameters; Adam adaptively scales updates using first and second moment estimates. In non-convex loss landscapes, optimization seeks a good basin rather than guaranteed global optimum.

B. Why ReLU Helps

ReLU is computationally simple and reduces vanishing-gradient behavior in positive regions.

C. Bias-Variance Perspective

Regularization, architecture constraints, and validation splits collectively manage the bias-variance tradeoff to avoid underfitting and overfitting extremes.

IX. Conclusion

The upgraded architecture is justified for heterogeneous tabular data, explainability is validated through SHAP, and theory is grounded in first principles. The final system is practical: model-tagged artifacts, threshold-aware evaluation, and decision-centric failure analysis make it submission-ready for applied AML/DL objectives.

References

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